Latent Render-and-Compare: SE(3) Diffusion with a Feature-Field World Model

Supplementary Material

This supplement gives implementation details (Sec. A), the fitting of the feature field (Sec. B), the evaluation protocols (Sec. E), our reproduction of LiePose and the choice between averaged and live weights (Sec. F), an audit of the information used at training and test time (Sec. G), and additional results (Sec. H). Section I describes the retrieval-referenced world model and reports all results obtained with it; its training procedure and geometry tokens are in Secs. C and D. Section, table and equation numbers without a letter refer to the main paper.

A. Implementation details

Observation encoding. Crops of 224×224 pixels are produced by LiePose's data pipeline, including its training-time augmentation (dynamic zoom-in, colour, blur and noise), and are exported once; DINOv2 ViT-S/14 (frozen, ImageNet normalisation) encodes them into 16×16 patch tokens that are average-pooled 2×2 to 64 tokens of 384 dimensions and stored in half precision. Foreground weights are the visible mask, max-pooled to the token grid and dilated by one token. On SPEED+, two augmented copies of every synthetic training image are exported; on T-LESS, 180,224 augmented training crops are drawn from LiePose's training set (PBR and real images). **[TODO: YCB-V numbers of exported crops.]**

Feature field. Table 1 lists the hyperparameters; the fitting procedure is in Sec. B. The field is evaluated in float32; the 32 depth samples per ray are at stratified random offsets while fitting and at bin centres in the denoiser and at test time.

Denoiser. Token transformer of width 256, depth 4, 8 heads, MLP ratio 4, adaLN modulation with zero-initialised gates in every block. Input tokens: a linear map of the concatenation of the separately LayerNorm-ed $\mathbf{z}_p$, $\hat{\mathbf{z}}_p$, $\hat{\mathbf{z}}_p - \mathbf{z}_p$ and $\mathbf{h}_p$, and w_p ($3 \cdot 384 + 128 + 1$ dimensions; control: $384+1$), plus a learned position embedding. The conditioning token is the sum of Fourier embeddings of the step and of the 6-dimensional Lie-algebra coordinates of the noisy pose, on SPEED+ plus a linear embedding of the crop features (log zoom, normalised crop centre, log box aspect), followed by SiLU and a linear layer. The output head maps the final conditioning token to the 6-dimensional noise. The field is applied with frozen parameters and a stop-gradient. With a trained encoder, the trainable DINOv2 copy (learning rate $\times 0.1$) yields 16×16 tokens; each 2×2 cell is concatenated to 1536 channels, LayerNorm-ed, projected to width 256 and added to the input tokens.

Diffusion, optimisation, sampling. The generative process is LiePose's SE(3) diffusion formulation, run with its public code and unchanged; it is not a contribution of this paper, and it is shared by LaRC, the control and the LiePose baseline, so that they differ only in the denoiser. As LiePose: SE(3) parametrisation with tangent-space representation, 100 noise levels with a variance-exploding power-law schedule, noise-prediction loss, Adam with learning rate 10^{-4} and exponential decay to 10^{-5} over the second half of training, weight averaging with decay 0.999 every 5 steps, and the 100-step sampler with the same options as LiePose's test script. The world model's parameters are not part of the optimised state. Where a batch of 64 images with 64 noisy poses does not fit a 40 GB GPU, the gradient is accumulated over 2 or 4 micro-batches; this gives the same gradient.

B. Fitting the feature field

Data. The field is fitted to the cached training tokens, the same crops that train the denoiser, at their ground-truth poses and with each crop's own crop camera. The appearance code of each crop is computed from its own tokens. On T-LESS a held-out 3% of the training source images is excluded for the discrimination measurements (Tab. 5).

Translation on T-LESS. LiePose's translation parametrisation encodes the offset of the annotated 2D centroid of the object from the crop centre. On about a quarter of the T-LESS training crops this differs from the projection of the object's 3D centre (by up to about 2 cm in x, y; depth is unaffected). The field is fitted at the true 3D centre, so that its geometry is consistent; the diffusion keeps LiePose's parametrisation, so at test time the field is queried at a hypothesis in that parametrisation.

Loss and optimisation. The weighted energy Eq. (2) at the ground-truth pose, Adam with gradient clipping at 1.0, learning rate $2 \cdot 10^{-3}$ with a linear warm-up over $\min(1000, \text{steps}/10)$ steps and cosine decay to 5%, batch 32. The density is never supervised.

Checkpoint. On SPEED+ the checkpoint with the highest 1–3° discrimination on the synthetic validation split is kept (step 60k, the last evaluation); on T-LESS the checkpoint with the lowest mean pick error on 500 held-out training crops, the criterion of the retrieval world model (Sec. C). Test data are not used.

Table 1. Hyperparameters. WM = world model. **[TODO: complete the YCB-V column.]**

	T-LESS	YCB-V	SPEED+	SYMSOL-T
Feature field (LaRC)				
tri-planes per object	$3 \times 128^2 \times 32$		$3 \times 128^2 \times 32$	–
MLP width / ray samples / rays	$128/32/16^2$		$128/32/16^2$	–
box half-size r_o	$1.2\times$ radius		1 m	–
fitting steps / batch / lr	$100\text{k}/32/2 \cdot 10^{-3}$		$60\text{k}/32/2 \cdot 10^{-3}$	–
token weights in the fit	mask (w_0=0.05)		uniform	–
field checkpoint	held-out pick error		val P_{1-3}	–
Retrieval world model (Sec. 1)				
WM width / depth / heads	384/6/6		384/6/6	384/6/6
WM geometry tokens	no		yes (geo) / no (ret)	yes
WM steps / batch	30k/32		30k/32	20k/32
WM optimiser	AdamW, lr $2 \cdot 10^{-4}$, wd 0.05, cosine, clip 1.0			
candidates (GT/uni/near/t/t-other)	1/11/12/4/4		1/11/12/4/4	1/11/12/4/4
near-rotation range	30°		30°	30°
σ_R / σ_t	10°/0.1		10°/0.1	15°/0.1
reference noise (rot.)	5°		5°	3°
retrieval λ	0		0	0.05
symmetries in bank	BOP models_info		none	analytic
WM checkpoint selection	held-out train images		synthetic val split	final step
denoiser width / depth / heads	256/4/8	256/4/8	256/4/8	256/4/8
steps $\times$ batch	400k$\times$64	400k$\times$64	400k$\times$64	400k$\times$32
noisy poses per image	64	64	64	64
LiePose (ours): steps $\times$ batch	400k$\times$64	400k$\times$64	400k$\times$64	800k$\times$16
LiePose (ours): noisy poses	256	256	256	256

C. Retrieval world model: training

Candidates. For each training crop with pose $g^\star = (R^\star, t^\star)$, a batch row contains: $g^\star$; 11 rotations drawn uniformly on SO(3) with $t^\star$; 12 rotations obtained by rotating $R^\star$ by a random axis and an angle up to 30°, with $t^\star$; 4 copies of $R^\star$ with $t^\star$ perturbed by Gaussian noise (scaled per axis to the range of the translation parametrisation); 4 copies of $R^\star$ with translations of other training crops. The rotation distance of each candidate to the ground truth is the minimum over the object's symmetric equivalents.

Reference. One reference per row, retrieved at a noisy copy of the ground truth and never from the row's own source image (exclusion is by source image, so augmented copies of the same image are excluded together). All candidates of the row share the reference, as the samples of one image share retrievals at nearby poses during diffusion.

Loss. Equation (2): cross-entropy between the softmax of negative energies over the candidates and the soft target built from the candidate distances, plus the energy at the ground truth. The temperature is a learned scalar initialised at e^{-2}.

SYMSOL-T extras. On SYMSOL-T the world model is trained from scratch on crops, and optionally with two sets of extra regression targets under the weighted-energy loss: in-plane rotations of training images (exact poses obtained by rotating the image), and pairs rendered with the SYMSOL-T renderer at small pose offsets from training poses. The rendered pairs are additional synthetic data that LiePose does not use (Sec. G). Four variants are trained for 30k steps (retrieval only; + geometry tokens; + rolls and pairs; + both), and one is selected on held-out training images (Sec. G); Tab. 10 reports all four. **[TODO: name the selected variant.]**

Checkpoint selection. T-LESS: 3% of the training source images are held out of the bank and of training; the checkpoint with the lowest selection error on 500 of them is kept, and test data are evaluated once. SPEED+: the same criterion on the synthetic validation split. SYMSOL-T: the final step of each variant, scored on held-out training images (Sec. G).

D. Retrieval world model: geometry tokens

Let K be the camera intrinsics and (s, x_0, y_0) the zoom and origin of a crop, so that a crop pixel is $s(u-(x_0, y_0))$. A pose

(R, t) in LiePose's parametrisation is mapped to the camera frame with the crop box (as in LiePose's translation recovery), including the conversion from allocentric to egocentric rotation. For the reference token p with crop-pixel centre c_p, we back-project its image position onto the plane $Z = Z^{\text{ref}}$ at the reference object depth, P_p, express it in the object frame, $o_p = (R^{\text{ref}})^\top (P_p - t^{\text{ref}})$, move it to the target pose, $P'_p = Ro_p + t$, and project it into the observation's crop. The token receives $[\, c_p/112 - 1,\ (\pi_k(P'_p) - c_p)/\Delta,\ \log(Z'_p/Z^{\text{ref}}) \,]$, with Δ the token size in pixels, Fourier-encoded at frequencies $2^{-2}, \ldots, 2^6$ and added through a zero-initialised MLP. The plane approximation is exact for the object centre and first-order for nearby surface points; it is only an input, never a metric. The seven virtual points are the object centre and the six points at $\pm r$ along the object axes, with r a nominal radius (half the shape size on SYMSOL-T; the largest keypoint radius on SPEED+); their projections at both poses, their image displacement and both depths form seven tokens with learned identity embeddings. Lens distortion is ignored.

E. Evaluation protocols

BOP (T-LESS, YCB-V). We use LiePose's test dataset on the BOP19 test targets and its evaluator, which writes the BOP CSV and calls the BOP toolkit. The toolkit's C++ renderer was not available in our environment, so VSD is computed with the toolkit's Python (vispy) renderer; this changes only how the depth maps used by VSD are rendered. MSSD and MSPD do not use a renderer. **[TODO: state the number of test instances per dataset.]**

SPEED+. Estimates are converted from LiePose's crop-relative parametrisation to the camera frame with the estimated translation and the crop box. For each image, $e_q = 2 \arccos |\langle q, \hat{q} \rangle|$ and $e_t = \|t - \hat{t}\|$. The score is $S_q + S_t$ with $S_q = e_q^{\text{rad}}$ set to zero when $e_q < 0.169°$ and $S_t = e_t/\|t\|$ set to zero when $e_t/\|t\| < 2.173$ mm/m, using the SPEC2021 thresholds [2]; each component is thresholded on its own, and the reported score, rotation part and translation part are means over all images of the lightbox and sunlamp sets. The SPEED+ paper [2] states the thresholds as zeroing the pose score only when both components are below them. **[TODO: check which rule the official SPEC2021 scoring code applies and use it for all rows; the two rules differ only for images with one component below its threshold.]**

SYMSOL-T. For each shape we report the mean over test images of the minimum rotation angle to the set of symmetric ground-truth rotations and of the Euclidean translation error, as LiePose. **[TODO: state the number of test images per shape used.]**

F. Reproducing LiePose and choosing weights

We retrained LiePose with its public code on every benchmark. On T-LESS our run uses batch 64 instead of the paper's 32 (400k steps in both cases), and our models are matched to our run. LiePose's test script samples with the exponential moving average (EMA) of the weights. For our T-LESS checkpoint, the EMA weights differ from the live weights by a small relative amount but give a much higher training loss than the live weights, and sampling with them fails; sampling with the live weights reproduces the published accuracy. We therefore choose, for every checkpoint we report (LiePose, control and LaRC, on every benchmark), between EMA and live weights by the training loss on fresh training batches, before any test evaluation. Table 2 lists the losses and the choices.

Table 2. Training loss of EMA and live weights on fresh training batches, and the weights used for testing. **[TODO: fill; values measured so far are recorded in the source comments. The control and LaRC numbers in the tables so far sample with EMA weights; the training-loss check for them has not been run yet.]**

Benchmark	Model	EMA	live	used
T-LESS	LiePose (ours)	5.39	2.50	live
T-LESS	Control			
T-LESS	LaRC			
YCB-V	LiePose (ours)			
YCB-V	Control			
YCB-V	LaRC			
SPEED+	LiePose (ours)	0.593	0.601	EMA
SPEED+	Control			
SPEED+	LaRC			
SYMSOL-T	LiePose (ours)			
SYMSOL-T	Control			
SYMSOL-T	LaRC			

G. Information audit

At test time. (i) Crops from ground-truth boxes on all benchmarks (Sec. 4.6): visible box and visible mask on T-LESS and YCB-V, as in LiePose; the box of the keypoints projected with the ground-truth pose on SPEED+; the box of the rendered foreground on SYMSOL-T, where LiePose uses the full image. (ii) The object identity, which selects the object's tri-planes. (iii) The crop camera, box and intrinsics, with which the field renders. (iv) The frozen field. No training image is needed at test time; with the retrieval world model (Sec. I), the tokens, poses and boxes of training crops are in its bank, and no test image is. No ground-truth pose is used by the model; ground truth is read only by the scorers and by rows and figures marked as diagnostics (oracle selection, oracle tokens, flip timing, world-model discrimination).

During training. Only training images: PBR and real training images on T-LESS and YCB-V, synthetic images on SPEED+ (the unlabeled lightbox and sunlamp images are not used in any form), and LiePose's training split on SYMSOL-T, plus the in-plane rotations of these images and the rendered pose pairs used by the SYMSOL-T world model (Sec. C).

Model selection. Denoisers: final checkpoint, EMA or live weights by training loss (Sec. F). World models: held-out training images on T-LESS, the synthetic validation split on SPEED+, and on SYMSOL-T 3% of the training images (1,350 per shape, none of them a source of the rolls or rendered pairs), which are removed from the world model's training set and reference bank. The selection score is the refinement pick error on 500 of these images (the energy's choice among 24 rotation perturbations of 0.25–8° around the ground truth); the variant with the lowest score at its final step is used, a rule fixed before training. Test images were not used to train or select any of the four variants. However, the variants themselves were designed during an earlier development phase in which the same diagnostic was computed on 500 test images; we disclose this, and Tab. 10 reports every variant on the test set.

Feature field. The denoiser input of LaRC, $[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w, \mathbf{h}]$, was chosen for the matched-budget runs after the six variants of Tab. 3 had been evaluated at 100k steps on the SPEED+ lightbox and sunlamp test sets; the choice is therefore informed by test results. All six variants are reported. The field's architecture and box sizes were fixed before any denoiser was trained.

Deviations from LiePose's setup. 64 instead of 256 noisy poses per training image for the control and LaRC; batch 64 for our T-LESS reproduction of LiePose (paper: 32); weights chosen by training loss (Sec. F); VSD rendered with the Python renderer. The control and LaRC train on a fixed set of pre-augmented crops whose tokens are cached (180,224 crops on T-LESS, two augmented copies of each synthetic image on SPEED+), whereas LiePose draws a new augmentation every time an image is used.

H. Additional results

Per-object results. **[TODO: per-object BOP recalls for T-LESS (30 objects) and YCB-V (21 objects), for LiePose (ours), the control and LaRC, as LiePose's Tab. 8.]**

Compute. **[TODO: GPU type and hours per run (world model, denoiser, LiePose), on the local workstation and the cluster.]**

Table 3. T-LESS rotation and translation accuracy (%): share of estimates with symmetric rotation error below 2/5/10 degrees and translation error below 2/5/10 cm, as reported by LiePose. **[TODO: fill.]**

Method	R@2	R@5	R@10	T@2	T@5	T@10
GDRNPP (det. boxes), from [1]	21.6	71.2	90.6	90.3	96.1	98.1
LiePose [1]	47.2	86.9	94.8	71.7	92.0	97.2
LiePose (our training)	50.5	86.7	92.3	74.1	91.7	96.8
Control (field), 100k	10.0	45.7	76.6	26.8	57.1	83.6
LaRC, 100k						
Control (ret.), 400k	11.3	47.1	76.4	34.8	69.5	88.2
LaRC (ret.), 400k	14.7	54.0	80.4	44.4	79.2	93.8

Table 4. SPEED+ breakdown: rotation and translation parts of the score, median errors, and the share of flipped estimates ($e_q > 90°$). **[TODO: fill.]**

Split	Method	S_q	S_t	med. e_q	med. e_t	flips
lightbox	LiePose (ours)	0.203	0.031	3.0	0.095	4.2%
	Control					
	LaRC (ret., geo.)	0.647	0.055	8.1	0.196	18.2%
	LaRC + sel.					
sunlamp	LiePose (ours)	0.260	0.029	5.0	0.107	5.5%
	Control					
	LaRC (ret., geo.)	0.669	0.047	9.0	0.189	19.7%
	LaRC + sel.					

Table 5. Training budget. The same denoiser at a short and at the matched budget (T-LESS AR %, SPEED+ lightbox S, SYMSOL-T mean rotation error), retrieval world model. **[TODO: fill; add the field rows once its 400k runs are tested.]**

Model	steps × batch	noisy poses	T-LESS	SPEED+	SYMSOL-T
Control	100k × 32	16	41.2	0.782	
LaRC	100k × 32	16	54.7	0.734	
Control	400k × 64 (32)	64	47.6		1.26
LaRC	400k × 64 (32)	64	54.3		1.01

Table 6. World-model design (discrimination as in Tab. 5, rotation perturbations 1–3° / 3–10°; selection error = rotation error of the lowest-energy candidate, deg). **[TODO: fill.]**

Variant (T-LESS)	1–3°	3–10°	selection error
Per-object learned template (no reference)	0.59	0.77	13.7
Retrieved reference, reconstruction loss only			
Retrieved reference, $\mathcal{L}_{\text{WM}}$	0.53	0.64	7.2
Feature field, reconstruction loss (LaRC)	0.53	0.66	14.2
without symmetric equivalents in the bank			
bank restricted to other objects (held-out objects)			

Broader impact. Pose estimation of known objects supports robotic manipulation and autonomous spacecraft rendezvous. We see no specific risk beyond those of perception systems in general; for safety-critical use, the failure modes in Fig. 4 and the reliance on ground-truth crops (Sec. 6) must be addressed first.

I. A retrieval-referenced world model

This section describes an alternative world model, which predicts the tokens at a pose from the nearest training view instead of rendering them from a field, and reports every result obtained with it. The denoiser, the control, the training budget and the evaluation are those of the main paper, except where stated. Everything in this section that says LaRC refers to LaRC with this world model.

I.1. Model

Reference bank and retrieval. For each object o, the *reference bank* holds the cached tokens, poses and crop boxes of the training crops, $\mathcal{B}_o = \{(\mathbf{z}_i, g_i, k_i)\}$. When the object's symmetry group $\mathcal{S}_o$ is known, the bank is expanded by it: each crop appears once per symmetric equivalent $(R_i S, t_i)$, $S \in \mathcal{S}_o$, subsampled to at most 36 (T-LESS) or 60 (SYMSOL-T) equivalents. Given a query pose $g = (R, t)$, retrieval returns the nearest entry under the key $[\mathrm{vec}(R_{\mathrm{allo}}),\ \lambda t]$, $\mathrm{ret}(g) = (\mathbf{z}^{\mathrm{ref}}, g^{\mathrm{ref}}, k^{\mathrm{ref}})$. While training both the world model and the denoiser, entries from the observation's own source image are excluded, so the model never sees its target as a reference. At test time the bank contains training crops only.

Prediction. The world model predicts the tokens of the object at pose g, in the crop k of the observation:

$$\hat{\mathbf{z}}(g) = \mathcal{F}_\phi^{\mathrm{ret}}\big(\mathbf{z}^{\mathrm{ref}},\ g^{\mathrm{ref}} \to g,\ k^{\mathrm{ref}},\ k\big) \in \mathbb{R}^{64\times 384}. \quad (1)$$

$\mathcal{F}_\phi^{\mathrm{ret}}$ is a 6-block transformer of width 384 whose blocks are modulated (adaLN [3]) by Fourier features of the target pose g and of the relative pose $g\,(g^{\mathrm{ref}})^{-1}$. Its input tokens are a linear projection of $\mathbf{z}^{\mathrm{ref}}$. The energy is Eq. (2) with this prediction and the reference retrieved at g.

Geometry tokens. A rotation of half a degree changes DINOv2 tokens very little, but it moves image points by a computable amount. When the camera intrinsics and the crop box are available, this displacement is given to the model explicitly (Sec. D). We use geometry tokens on SYMSOL-T and SPEED+; on T-LESS the token cache used for this model does not store the per-crop intrinsics.

Training. The world model is trained so that its energy ranks poses, not only so that it reconstructs tokens. For each training crop with pose $g^\star$ we form a candidate set $\mathcal{C}$ (Sec. C). The reference is retrieved at a noisy copy of $g^\star$, which mimics the poses at which the denoiser will query it. With a soft target that accounts for symmetry, $q_c \propto \exp\big(-d_{\mathcal{S}}(R_c, R^\star)^2/2\sigma_R^2 - \|t_c - t^\star\|^2/2\sigma_t^2\big)$, where $d_{\mathcal{S}}$ is the rotation distance to the nearest symmetric equivalent, the loss is

$$\mathcal{L}_{\mathrm{WM}} = \mathrm{CE}\Big(\mathrm{softmax}_c\big(-E(g_c)/\tau\big),\ q\Big) + \lambda_{\mathrm{rec}}\, E(g^\star), \quad (2)$$

with a learned temperature τ. On SYMSOL-T we add regression targets from in-plane rotations of training images and from rendered pose pairs (Sec. C).

Denoiser. The denoiser's input tokens are $[\mathbf{z}_p,\ \hat{\mathbf{z}}_p,\ \hat{\mathbf{z}}_p - \mathbf{z}_p,\ w_p]$, and the control sets $\hat{\mathbf{z}} := 0$. The conditioning includes a linear embedding of the crop box on all benchmarks. Retrieval is a nearest-neighbour query on the CPU, once per sample and step.

Differences from the field. The retrieved prediction carries the lighting, background and occlusion of the reference view, and its error at the true pose is bounded by how far the nearest training view is. The field has neither limitation, but it must represent the whole object from tokens alone, and it needs no reference bank at test time.

I.2. Results

Summary. On T-LESS the retrieval world model raises AR over its control by 13.5 and 11.6 points at a quarter of the budget (seeds 42 and 43) and by 6.7 at the matched budget. The rendered tokens carry the gain: the explicit difference and the weight add 0.9 and 0.1 points over $[\mathbf{z}, \hat{\mathbf{z}}]$. The comparison contributes at low noise (52.4) and not at high noise (40.7, no better than the control). On SPEED+ the gains are smaller (0.782 to 0.734 at 100k; 0.717 with geometry tokens), and the 400k model with geometry tokens (0.701) is far behind LiePose (0.234). With a trained encoder, the comparison lowers the score at the intermediate checkpoint from 0.306 to 0.280, and from 0.204 to 0.168 with consensus selection. On SYMSOL-T it lowers the mean rotation error on every shape. Objects never seen by the denoiser remain out of reach: AR 6.3 against 3.5 for the control.

References

[1] Tsu-Ching Hsiao, Hao-Wei Chen, Hsuan-Kung Yang, and Chun-Yi Lee. Confronting ambiguity in 6D object pose estimation via score-based diffusion on SE(3). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 4, 6

[2] Tae Ha Park, Marcus Märtens, Gurvan Lecuyer, Dario Izzo, and Simone D'Amico. SPEED+: Next-generation dataset for spacecraft pose estimation across domain gap. In *IEEE Aerospace Conference*, 2022. 3

[3] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 5

Table 7. **T-LESS with the retrieval world model** (BOP19 targets, BOP average recall, %, ground-truth crops). Protocol as Tab. 1. ‡Encoder trained (a frozen copy gives the comparison space).

Method	Encoder	Budget	Seed	VSD	MSSD	MSPD	AR
LiePose (our training)	ResNet-34	400k × 64	42	63.1	65.7	92.3	73.7
Control (no world model)	DINOv2-S	400k × 64	42	30.5	34.0	78.3	47.6
LaRC, retrieval	DINOv2-S	400k × 64	42	38.4	42.3	82.4	54.3
Control (no world model)	DINOv2-S	100k × 32	42	22.2	25.8	75.6	41.2
LaRC, retrieval	DINOv2-S	100k × 32	42	38.2	42.5	83.3	54.7
Control (no world model)	DINOv2-S	100k × 32	43	23.4	27.1	76.9	42.5
LaRC, retrieval	DINOv2-S	100k × 32	43	37.8	42.0	82.6	54.1
Control, trained encoder	DINOv2-S‡	100k × 32	42				
LaRC, retrieval, trained encoder	DINOv2-S‡	100k × 32	42				

Table 8. **SPEED+ with the retrieval world model** (protocol as Tab. 2). ‡Encoder trained.

			lightbox			sunlamp		
Method	Box	Samples	e_t	e_q	S	e_t	e_q	S
Matched number of training images (400k × 64)								
Control (no world model)	GT	1						
LaRC, retrieval	GT	1						
LaRC, retrieval + geometry tokens	GT	1	0.317	37.0	0.701	0.275	38.3	0.716
Trained encoder‡; intermediate checkpoint (100k of the 400k schedule)								
Control (no world model)	GT	1	0.159	16.0	0.306	0.215	21.8	0.417
+ consensus	GT	16	0.144	10.3	0.204	0.202	16.1	0.314
LaRC, retrieval	GT	1	0.154	14.5	0.280	0.204	19.7	0.379
+ consensus	GT	16	0.128	8.4	0.168	0.184	12.9	0.255
Trained encoder‡, final checkpoint (400k × 64)								
Control (no world model)	GT	1						
LaRC, retrieval	GT	1						

Table 9. **SYMSOL-T with the retrieval world model** (mean rotation error R in degrees, mean translation error t; lower is better). *Input*: full image or crop around the rendered object. Our rows: 400k × 32, 64 noisy poses. On SYMSOL-T LiePose takes the full image, so the crop gives LaRC information that LiePose does not use. **[TODO: fill the LiePose rows.]**

		tet.		cube		icosa.		cone		cyl.	
Method	Input	R	t	R	t	R	t	R	t	R	t
Regr. [1]	full	2.92	0.064	2.86	0.050	2.46	0.037	1.84	0.058	2.24	0.049
Iter. regr. [1]	full	4.25	0.048	4.20	0.037	29.33	0.026	1.63	0.037	2.34	0.032
LiePose [1]	full	0.59	0.016	0.58	0.011	0.64	0.012	0.54	0.016	0.41	0.011
LiePose (ours)	full										
LiePose (ours)	crop										
Control	crop	1.72	0.048	1.03	0.028	0.84	0.022	1.90	0.045	0.79	0.031
LaRC (ret.)	crop	1.35	0.039	0.90	0.022	0.57	0.016	1.53	0.032	0.70	0.020

Table 10. **Retrieval world-model variants.** Same denoiser and budget (100k steps × batch 32, 16 noisy poses per image), one sample. T-LESS: AR (%, ↑); SPEED+: score S on lightbox (↓); SYMSOL-T: mean rotation error over shapes (deg, ↓).

World model in the denoiser	T-LESS	SPEED+	SYMSOL-T
None (control, $\hat{\mathbf{z}}$=0)	41.2	0.782	
Learned template, no reference			
Retrieved reference	54.7	0.734	
+ geometry tokens	–	0.717	
+ roll / render-pair targets	–	–	
Oracle $\hat{\mathbf{z}}$: GT-view tokens (diag.)			

Table 11. **Denoiser inputs and when the comparison is used**, retrieval world model. T-LESS AR (%, ↑) and SPEED+ lightbox S (↓), one sample, 100k × 32 (16 noisy poses per image) for every row. $t_0 = 50$ of the 100 training timesteps, fixed in advance; outside the stated range the prediction is zeroed, as in the control.

Variant	T-LESS	SPEED+
$[\mathbf{z}]$ (control)	41.2	0.782
$[\mathbf{z}, \hat{\mathbf{z}}]$	53.7	0.716
$[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}]$	54.6	0.718
$[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w]$ (LaRC)	54.7	0.734
Comparison only for $t \geq t_0$ (high noise)	40.7	0.774
Comparison only for $t < t_0$ (low noise)	52.4	0.755
16×16 full-resolution tokens		

Table 12. **Selection with the retrieval world model** (SPEED+ lightbox S, ↓), the 100k model with geometry tokens (Tab. 10). Random: sample 0; oracle: the sample closest to the ground truth in rotation (diagnostic).

N	random	energy	oracle
1	0.713	0.713	0.713
16	0.713	0.593	0.331

Table 13. **Held-out objects** (T-LESS, BOP AR %, retrieval world model, 100k × 32). The denoiser is trained on objects 1–20 only and tested on all 30; at test time each object's bank holds its training crops. **[TODO: state which objects the world model was trained on (`tless_ret_ho20`).]**

Method	all	seen (1–20)	unseen (21–30)
Control (no world model)	34.7	44.4	3.5
LaRC (ret.)	41.4	52.4	6.3